

NIAGARA INTAKE, NY, USA

WMO: 998235

Lat: 43.0770N

Lon: 79.0140W

Elev: 587

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.0	(c) 8.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.6	(c) 84.1	(d) N/A	(e) 81.9	(f) N/A	(g) 79.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	0.7	88.8	5.2	2.1	-3.1	90.3	-6.2	91.5	-9.1	92.7	-12.9	94.3	
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	24.5	26.0	34.1	45.6	58.3	66.3	72.3	71.0	64.6	53.2	41.6	33.0		
	DBStd	18.33	9.98	9.56	9.55	7.18	7.56	5.36	4.85	4.08	6.43	6.90	8.31	8.58		
	HDD50	2998	792	674	500	168	17	0	0	0	0	45	273	529		
	HDD65	6322	1256	1093	958	582	230	48	4	83	371	701	992			
	CDD50	2760	1	1	8	37	276	490	693	650	438	143	21	2		
	CDD65	607	0	0	0	0	24	88	231	189	71	4	0	0		
	CDH74	2906	0	0	1	3	135	339	1330	783	306	9	0	0		
	CDH80	413	0	0	0	0	7	31	249	88	38	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	33.6	2.7	2.2	2.4	3.0	3.0	3.1	3.0	3.3	3.1	2.7	2.8			
	PrecMax	43.5	4.6	4.7	4.2	5.1	6.3	7.0	7.4	5.5	7.5	6.9	4.8	6.1		
	PrecMin	26.9	1.5	0.9	0.7	1.3	0.8	0.8	0.7	1.3	1.5	1.1	1.1	1.1		
	PrecStd	4.8	0.9	0.9	1.0	1.1	1.5	1.5	1.6	1.1	1.4	1.4	0.9	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.4	54.9	69.0	73.5	80.8	83.4	87.9	85.6	83.9	75.3	65.3	59.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	47.9	48.0	57.7	67.4	77.9	80.4	85.5	82.3	80.8	70.2	61.2	52.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	43.0	42.3	52.7	62.4	74.9	77.9	82.9	80.3	77.5	66.4	57.5	49.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	38.4	38.0	47.9	58.3	71.4	75.4	80.4	78.3	75.0	63.5	54.4	45.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		11.0	10.8	12.8	15.5	16.4	13.7	12.6	11.8	13.0	11.4	10.9	9.5		
	5% DB	MCDBR	16.0	17.2	20.0	23.5	20.3	16.6	14.9	13.7	14.7	15.7	15.4	13.3		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.297	0.324	0.334	0.374	0.404	0.434	0.433	0.421	0.380	0.361	0.314	0.298		
	taud		2.414	2.331	2.404	2.329	2.289	2.239	2.258	2.281	2.374	2.409	2.507	2.468		
	Ebn at Noon		266	275	288	283	276	267	266	265	268	259	259	256		
	Edh at Noon		24	31	33	38	41	43	42	40	34	29	22	21		
(44) All-Sky Solar Radiation	RadAvg		418	668	1047	1400	1718	1854	1867	1656	1318	796	494	338		
	RadStd		39	62	113	149	180	140	143	89	118	74	68	43		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page